

Optimising and Evaluating Neural Networks for Classifying Paediatric Hyperkinetic Movement Disorders

6CCYB070 BEng Research Project



George Welling

Supervisor: Dr Rachel Sparks
Dr Daniel Lumsden

School of Biomedical Engineering and Imaging Sciences
King's College London

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Project Plan

Problem

Hyperkinetic movement disorders (HMDs) in children are difficult to diagnose due to their reliance on subjective clinical assessment. HMDs vary widely in severity and often present with overlapping or mixed features, meaning that in clinical diagnosis it is not strictly a binary problem. These challenges are further reinforced by the fact that early motor development can be highly variable, and many symptoms are dismissed as delayed maturation. HMDs affect an estimated 1 in 100 children, yet diagnosis can take several years, delaying access to appropriate treatment.

Previous work developed a spatio-temporal neural network demonstrating strong performance in classifying two common HMDs: chorea and dystonia. The relative contribution of each network component to the overall performance was not evaluated. Additionally, the integration of a simple self-attention mechanism within the recurrent layers potentially limits model interpretability and may lack temporal consistency between features.

This project aims to systematically evaluate the contribution of network components via an ablation study, evaluate data normalisation methods to ensure a standardised scaled representation of features and explore different attention mechanism designs to enhance temporal feature learning.

State of the Art

Current clinical evaluation of paediatric HMDs primarily depends on expert clinical interpretation, with high inter-observer variability and inconsistent diagnostic accuracy, especially among non-expert neurologists. Recent studies have explored deep learning techniques, including convolutional and graph-based neural networks, to classify movements such as dystonia, chorea and tremor. However, existing models are not easy to interpret, depend on small and highly curated datasets, and lack systematic evaluation. As a result, no clinically deployable pipeline exists, and clinicians continue to rely on generalised assessment strategies shaped by experience rather than quantitative evidence.

Aims

This study aims to systematically evaluate the performance of a previously presented spatio-temporal network, assessing the respective contributions of the spatial and temporal components, the impact of techniques for model robustness (dropout, data normalisation), and the influence of attention mechanisms on the model performance. The results of this work will lead to improved model stability and classification accuracy demonstrated in the context of two HMDs: chorea and dystonia.

Work Plan

This project is divided into three tasks.

Task 1 will conduct an ablation study on the spatio-temporal network to analyse the relative contributions of the spatial and temporal modules on classification accuracy. As a part of this analysis, different parameters including dropout rate, total number of training epochs and learning rate will be tuned in a grid search.

Task 2 will develop a data normalisation method to generalise the model to data collected from video cameras with different aspect ratios. The relative size of the height and width of keypoints will be used to normalise videos taken from different cameras and patients.

Task 3 will evaluate different attention mechanism implementations for the recurrent layers within the temporal modules. This includes reviewing single-headed and multi-headed attention formulations, potentially adding normalisation for feature stability, and addressing any training instabilities that may be observed.

Deliverables

This project aims to deliver an improved deep learning framework for the classification of HMDs in the paediatric population. The outcome will include a refined and characterised model architecture, improvements to spatial and temporal normalisation methods, and an integrated attention mechanism with clearly justified design choices.

Evaluation

We will divide the entire dataset into training, validation and testing subsets. Model performance will be evaluated on an unseen test set to ensure generalisability. Given the classification focus of this project, performance will be assessed using accuracy, sensitivity, specificity and F1 score, with the objective of reducing misclassification while maintaining strong separation between HMD categories.

Project Timeline

	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26
Task 1							
Task 2							
Task 3							
Report							

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Abstract

Hyperkinetic movement disorders (HMDs) in paediatric populations, particularly dystonia and chorea, are challenging to diagnose due to their reliance on subjective clinical assessment and the heterogeneity of movement presentations. This study investigates the optimisation and interpretability of a pose-based spatio-temporal neural network for automated classification of these disorders using video-derived skeletal keypoints.

A deep learning pipeline was implemented in which clinical video recordings were converted into structured skeletal representations using pose estimation, enabling modelling of both spatial joint relationships and temporal movement dynamics. The baseline model, combining a graph-based spatial module with a bidirectional LSTM temporal module and self-attention mechanisms, was systematically evaluated through architectural and preprocessing ablation studies. These included removal or modification of network components, comparison with alternative temporal architectures and assessment of preprocessing strategies such as spatial normalisation, data augmentation and frame rate variation.

The baseline model achieved strong classification performance, with a balanced accuracy of 85.9% and robust sensitivity and specificity across both classes. Ablation results demonstrated that both spatial and temporal components are critical for accurate classification, with significant performance degradation observed when either was removed. Temporal attention mechanisms were also shown to substantially improve performance by enabling the model to focus on diagnostically relevant segments of motion. Preprocessing techniques, particularly spatial normalisation and data augmentation, were found to enhance model robustness and generalisation.

Interpretability analysis using attention maps, saliency maps and Grad-CAM revealed distinct temporal activation patterns corresponding to clinical characteristics of dystonia (sustained movements) and chorea (intermittent movements), suggesting that the model captures meaningful physiological features.

Despite limitations including small dataset size and binary classification constraints, this study demonstrates that spatio-temporal deep learning approaches can provide accurate and interpretable classification of paediatric HMDs. These findings support the potential for developing objective, video-based clinical decision-support tools to improve diagnostic consistency and accessibility in movement disorder assessment.

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1 Introduction

1.1 Clinical Problem

1.1.1 Hyperkinetic Movement Disorders: Overview & Prevalence

Movement disorders are classified as hyperkinetic and hypokinetic. Hyperkinetic movement disorders (HMDs) are characterised by excessive, involuntary movement, whereas hypokinetic disorders involve reduced movement [1]. In paediatric populations, hyperkinetic disorders are common and include dystonia, chorea, tics, tremor, myoclonus and stereotypies [2, 3].

Hyperkinetic movements represent abnormal motor outputs resulting from impaired modulation of voluntary movement. Under normal physiological conditions, motor activity is regulated by complex circuits involving the basal ganglia, thalamus, cerebellum and cerebral cortex, specifically the sensorimotor circuit, shown in Figure 1 [4]. Disruption within these networks, whether due to static injury, progressive neurodegeneration, autoimmune processes, metabolic disease or genetic mutation, can lead to disinhibition or dysregulation of motor pathways. Clinically, this is seen as involuntary, excessive movements. [1, 3]. Increasingly, HMDs are understood not as disorders of a single anatomical structure but as a dysfunction of distributed motor networks.

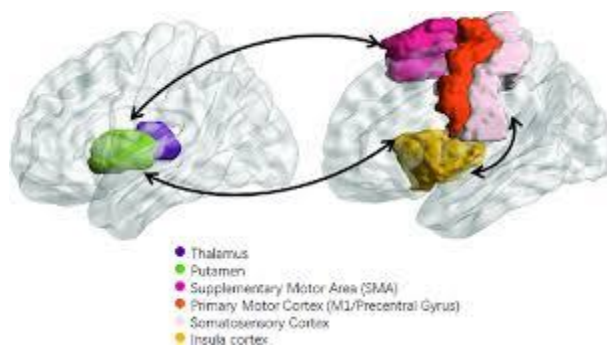


Figure 1: Image of the Sensorimotor Circuit, showing affected areas in patients with HMDs

In children, HMDs constitute a significant proportion of referrals to paediatric neurology services [2]. They may present as primary (often genetic) or secondary to conditions such as cerebral palsy, encephalitis or metabolic disease [2, 3]. The heterogeneity in aetiology contributes to substantial variability in prevalence estimates; however, hyperkinetic phenotypes are recognised as a major contributor to childhood motor disability worldwide [1, 2]. The conditions also affect functional independence, communication and quality of life [5, 6].

Given the broad spectrum of hyperkinetic presentations, accurate clinical characterisation is essential. Among the most clinically significant and frequently encountered hyperkinetic syndromes in paediatrics are dystonia and chorea.

Dystonia

Dystonia is characterised by sustained or intermittent involuntary muscle contractions causing abnormal movements and postures [3]. These contractions typically involve co-contraction of agonist and antagonist muscle groups, producing twisting movements or fixed abnormal postures. Dystonia is dynamic in nature: symptoms may fluctuate in severity and are often exacerbated by voluntary movement, stress or fatigue [3]. Clinically, dystonia may be focal, segmental, hemidystonic or generalised [1].

Historically, dystonia was conceptualised primarily as a basal ganglia disorder. Lesions within the putamen and globus pallidus are strongly associated with dystonic symptoms [7]. However, contemporary understanding emphasises dystonia as a network disorder involving abnormal connectivity between the basal ganglia, cerebellum, thalamus and cerebral cortex [7].

In paediatric populations, primary (isolated) dystonia is frequently genetic, whereas secondary dystonia commonly arises from perinatal brain injury, metabolic disease, infection or neurodegeneration [2, 3]. Dystonia can be severely disabling, particularly when generalised, and often requires multidisciplinary management including pharmacological therapy, botulinum toxin injections or neurosurgical interventions such as deep brain stimulation (DBS) in selected cases [8].

Chorea

Chorea consists of irregular, non-rhythmic, involuntary movements that flow between body regions [3, 9]. Unlike dystonia, chorea does not typically produce sustained postures. Instead, movements are fragmented and variable. Because they may be partially incorporated into voluntary movements, choreiform movements can give the impression of restlessness, complicating clinical recognition [3].

Pathophysiologically, chorea is associated with dysfunction of cortico-striato-thalamo-cortical circuits, particularly involving the basal ganglia [1, 9]. However, similar to dystonia, increasing evidence supports involvement of broader motor networks including the cortex, thalamus and cerebellum [3]. The specific clinical phenotype of chorea may vary depending on its underlying cause.

In paediatrics, chorea may arise from post-infectious autoimmune conditions (such as streptococcal-associated chorea), encephalitis, metabolic disorders, structural brain injury or genetic diseases [1, 2, 3]. Distinct aetiologies often produce differing symptom patterns, associated neurological signs and disease trajectories. For example, post-streptococcal chorea may present with emotional lability and

behavioural changes, whereas chorea associated with encephalitis may occur alongside seizures or cognitive impairment [3].

Clinical Significance

Although dystonia and chorea are mechanistically and phenomenologically distinct, both represent disorders of impaired motor circuit regulation. Their overlapping anatomical substrates and heterogeneous causes highlight the importance of brain network-based approaches to understanding HMDs. In paediatric populations particularly, early recognition and accurate classification are essential to guide appropriate investigation, management and long-term prognosis.

1.1.2 Current Clinical Diagnosis

Dyskinetic cerebral palsy is a subtype of cerebral palsy (CP) characterised by involuntary movements, fluctuating muscle tone and abnormal postures. It is typically classified into two predominant motor phenotypes: dystonic and choreoathetoid subgroups [10]. Accurate differentiation between these subtypes is clinically important, as symptom presentation, progression and management strategies differ.

Assessment of dystonia severity in clinical practice most commonly relies on standardised rating scales. The Barry-Albright Dystonia Scale (BADS) and the Burke-Fahn-Marsden Dystonia Rating Scale (BFMDRS) are the most frequently reported outcome measures [10]. BADS evaluates dystonia severity across eight body regions, whereas BFMDRS assesses in nine regions and includes both movement and disability components.

Despite the availability of these structured tools, clinical differentiation between dystonia and chorea remains dependent on subjective interpretation by trained clinicians. Accurate classification requires clinicians to distinguish subtle phenomenological differences, which can be challenging due to overlapping motor features resulting in higher intra-observer variability between clinicians.

1.1.3 Deep Learning for Video-Based Clinical Movement Analysis

Machine learning (ML) is a subset of Artificial Intelligence, encompassing a broad range of algorithms that perform predictions based on previously seen data, referred to as the training set [11]. ML is broadly divided into supervised and unsupervised learning. In supervised learning, models are trained on data with associated labels, and the ML model adjusts the internal parameters to minimise prediction error of the labels and learn mappings between input data and output labels [11, 12]. In contrast, unsupervised learning uses unlabelled data, and ML models identify underlying patterns or groupings within the data, commonly through clustering measures [11, 12]. Deep learning (DL) is a

specific subclass of ML models that use multiple layers of interconnected neurons (hidden layers), enabling the processing and analysis of more complex relationships between the data [12].

Over the past decade, machine learning has transformed healthcare data analysis. While traditional models rely on manually engineered features, deep learning automatically learns hierarchical representations from unstructured data. This has enabled advances in medical imaging and clinical decision support. In neurological disorders, where phenotypes are complex and temporally dynamic, deep learning provides a framework for identifying patterns not readily observable through clinical assessment.

Unlike traditional approaches that rely on handcrafted features, deep neural networks can extract meaningful patterns from high dimensional healthcare data, including medical imaging, electronic health records and physiological signals. Deep learning has driven significant advances in automated prediction, diagnosis and phenotyping across multiple clinical domains, while also introducing challenges related to model and output interpretability, data quality and clinical integration [13].

Many neurological conditions, including Parkinson's disease and HMDs, are characterised by complex movement patterns that cannot be fully captured through static imaging or clinical observations. Video recordings provide a valuable modality for assessing temporal motor behaviours, allowing clinicians to observe movement and movement abnormalities. However, the current clinical standard of care is manual interpretation of these recordings that remains subjective and time consuming, similar to in person examinations. A major contribution to the subjectiveness of in person examinations, is the difference in expert observer examination techniques which can lead to different conclusions.

Deep learning has been applied to both spatial representations (i.e. pose detection) and temporal representations (i.e. time series forecasting). For tasks related to human actions, models combining both the spatial and temporal components have been presented and are called spatio-temporal architectures to indicate the model can characterise both types of features. These techniques have been successfully applied to the analysis of neurological movement disorders, enabling automated assessment of gait abnormalities and motor impairment directly from standard video recordings without the need for specialised motion capture systems [14]. These approaches can capture subtle movement characteristics that may be difficult to quantify through traditional clinical observation alone, supporting more objective and scalable assessment of motor dysfunction.

Building on these developments recent work has explored the use of pose-based spatio-temporal neural networks to classify paediatric HMDs using video-derived skeletal keypoints [15]. By modelling both the spatial relationships between body joints and their temporal evolution across frames, these systems demonstrate the potential of deep learning to distinguish between different movement phenotypes such as dystonia and chorea.

1.2 State of the Art

1.2.1 Pose Estimation Techniques

Human pose estimation is a computer vision technique that detects anatomical keypoints from images or video, enabling a reconstruction of a skeletal representation of human movement. Modern deep learning-based approaches use convolutional neural networks to determine joint locations, typically via heatmap prediction, and are commonly implemented using top-down (person-first) or bottom-up (joint-first) frameworks. Architectures such as OpenPose, HRNet and related high-resolution networks have demonstrated high accuracy in multi-person 2D pose estimation [16, 17]. Figure 2 shows the use of the OpenPose Body25 Model, demonstrating where the 25 keypoints are on a human body [18].

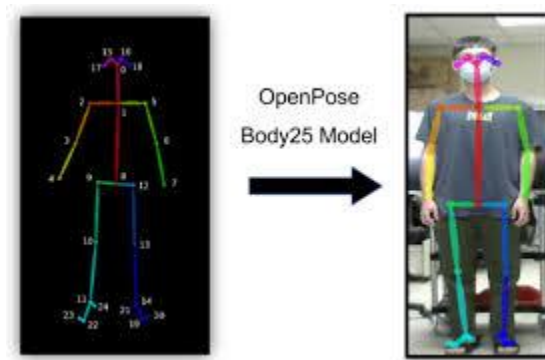


Figure 2: Diagram demonstrating OpenPose Body25 Model Keypoints

Pose estimation proves an efficient intermediate representation for movement analysis by converting high-dimensional video data into structured skeletal coordinates describing body joint positions. This representation significantly reduces data complexity while preserving essential biomechanical information about posture and motion. Pose-based representations can enhance data privacy in clinical settings, as skeletal keypoints remove identifiable visual information while retaining the movement characteristics required for computational analysis [19, 20].

For video-based analysis, temporal modelling is often incorporated to exploit motion continuity over frames. Temporal convolutional networks and attention-based architectures allow models to capture sequential dependencies, improving robustness to noise and transient misdetections. However, temporal performance is sensitive to frame rate and video quality. In clinical environments, low frame-rate recordings may inadequately capture rapid hyperkinetic movements, limiting temporal feature extraction and potentially reducing classification reliability [17, 21, 22].

1.2.2 The Role of Ablation in Model Optimisation and Interpretation

As deep neural networks have grown in architectural complexity, systematic evaluation of component-wise contributions has become essential [23]. Ablation analysis refers to the structural removal,

modification or isolation of model components to quantify their impact on predictive performance. Within deep learning research, ablation studies are routinely used to validate architectural design choices and to ensure that performance gains arise from meaningful structural innovations rather than increased number of parameters (which is an indication of model complexity) or artefacts arising from training or input data [24, 25].

Formally, ablation can be framed as evaluating the performance differential:

$$\Delta\mathcal{L} = \mathcal{L}_{full} - \mathcal{L}_{ablated},$$

where $\mathcal{L}$ represents a task-specific loss or evaluation metric. By measuring changes in performance after removing a given module, its contribution to model behaviour can be approximated [26]. This approach is particularly important in high-capacity models where interactions between components may not be intuitively interpretable.

Ablation strategies can be broadly categorised into architectural ablation and feature or parameter ablation. Architectural ablation evaluates the necessity of structural components such as temporal convolutional layers, residual connections or attention mechanisms. This form of analysis is widely employed in computer vision and transformer-based models to justify decisions related to model design [27, 28]. In contrast, feature ablation selectively removes input variables, spatial keypoints or temporal segments to assess their relative importance. Within pose-based movement analysis, feature ablation can reveal which anatomical joint or temporal windows contribute most strongly to the model output.

Interpretability is critical in the deployment of deep learning systems within clinical environments [23]. High predictive performance for models is desirable; however, clinicians must also be able to understand and trust the reasoning of underlying models, particularly in diagnostic contexts where decisions may influence patient management. Black-box behaviour in neural networks has therefore become a major concern in medical AI research, motivating the development of explainable artificial intelligence (XAI) techniques that reveal which regions of the input data or which internal features most strongly influence predictions [29, 30]. Providing interpretable evidence of model reasoning can increase clinical confidence, support error analysis and help identify potential bias or spurious correlations in the learned representations [30, 31].

Several visualisation techniques have been proposed to improve interpretability in deep learning models. Gradient-based attribution methods such as saliency maps highlight regions that most strongly affect the output by computing the gradient of the prediction with respect to the input features [32]. Similarly, Gradient-weighted Class Activation Mapping (Grad-CAM) produces spatial heatmaps indicating which image regions contribute most strongly to a specific classification decision and has been widely applied in medical imaging tasks to provide visual explanations for convolutional neural

network predictions [33]. In architectures incorporating attention mechanisms, attention maps can provide additional insight by revealing how much the model allocates importance across spatial features or temporal segments during inference. Comparing these visualisation approaches can help determine whether attention mechanisms align with gradient-based explanations, providing further evidence of model reliability and interpretability [25].

1.3 Clinical Data Complexity

Clinical movement data are typically collected in heterogeneous, unconstrained environments, with variability in camera placement, background noise and occlusion. This introduces shifts in data distribution across sessions, patients and recording sites [34, 35]. As a result, data preprocessing steps such as spatial normalisation and standardisation of pose representations may be required to reduce inter-recording variance and ensure that models learn clinically meaningful motion patterns rather than information contained in the video related to differences in recording setups.

Robustness is therefore essential for clinical deployment. Models must maintain performance across demographic variation, recording conditions and disease heterogeneity, as performance degradation directly impacts diagnostic reliability. [30].

Dataset size presents an additional limit. Clinical cohorts are often small due to recruitment and ethical constraints, increasing the risk of overfitting, particularly in high-capacity models [36]. Techniques such as data augmentation and regularisation are therefore used to improve generalisation.

Noise and missing data further complicate modelling. Pose estimation is susceptible to keypoint errors, occlusion and tracking discontinuities, introducing structured noise. Missing data may also arise from occlusion or variable recording lengths and may correlate with disease characteristics, increasing the risk of learning spurious associations [37].

These challenges differ substantially from those encountered in conventional human action recognition tasks. Large-scale action recognition datasets such as Kinetics or UCF101 are typically constructed from publicly available internet videos, enabling the collection of thousands of different labelled samples across many action categories [38, 39]. In contrast, clinical datasets require patient consent, ethical approval and specialist annotation, making large-scale data collection significantly more difficult. Furthermore, clinical movements often exhibit subtle heterogeneous phenotypes rather than clearly defined actions, increasing the difficulty of automated classification.

Crucially, many of these challenges are intrinsic to clinical data and persist regardless of model choice. Increasing architectural complexity does not eliminate distributional shift, limited sample size or structured noise. Instead, systematic model evaluation becomes necessary to determine which components improve generalisation and which amplify sensitivity to artefacts or noise in the data.

Within this context, ablation analysis provides a principled mechanism for determining model robustness. By selectively removing architectural modules or input features, one can assess whether performance depends on clinically meaningful signals or unstable data characteristics [40]. This establishes a direct methodological link between clinical data complexity and the need for structured ablation model testing within the present study.

1.4 Objectives

The fundamental aim of this project is to optimise and systematically characterise a spatio-temporal neural network for the classification of paediatric hyperkinetic movement disorders (HMDs), specifically dystonia-predominant and chorea-predominant cases. The focus is not only on maximising classification performance, but on understanding the relative contribution of architectural components and improving model robustness within realistic clinical constraints.

The main objectives are:

- To perform an ablation study of spatial and temporal components,
- Implement spatial normalisation for cross-environment robustness,
- Evaluate the temporal attention mechanisms using model interpretability techniques,
- Validate performance on unseen data using clinically relevant measurements.

2 Methodology

2.1 Study Overview

This study investigates the optimisation and interpretability of a spatio-temporal neural network for the classification of paediatric hyperkinetic movement disorders, specifically dystonia-predominant and chorea-predominant cases. The data processing pipeline follows a video-based movement analysis framework in which raw clinical recordings are first converted into skeletal representation using a pose estimation software. These skeletal keypoints provide a structured representation of body joint positions over time, enabling the modelling of both spatial relationships between joints and temporal patterns of joints over time.

The underlying model architecture is based on the spatio-temporal neural network presented in prior work by Nandika et al. [15], which combines a spatial graph neural network representation of body joints within a single frame with a temporal recurrent neural network to capture movement dynamics across frames. Building on this foundation, the present study focuses on systematic evaluation of the model through architectural and preprocessing ablation studies, and detailed analysis of the temporal recurrent neural network behaviour using different model interpretation techniques. Gradient-based visualisation methods, attention and saliency maps are employed to examine how temporal information contributes to classification decision of the model.

2.2 Dataset and Data Preparation

2.2.1 Dataset Description

This study used videos collected from the routine assessment of children and young people selected for Deep Brain Stimulation (DBS) at the Evalina Children’s Hospital (ECH) under the care of the Complex Motor Disorder Service. Regulatory approval was granted by the Guy’s and St Thomas’ NHS Foundation Trust (GSTT) Electronic Records Research Interface (GERRI) (REC# 20EM/0112), under the supervision of the Children’s Neurosciences department. The video records of the most recent 100 children and young people undergoing DBS prior to 1 December 2023 were available for analysis under this approval. Videos recorded as part of the standard BFMDRS protocol were used as the data source for this project. Video recordings were reviewed by Dr. Lumsden, who has over 15 years’ experience in the management of disorders in childhood. Each recording was clinically labelled as either dystonia-predominant or chorea-predominant by Dr. Lumsden with review by a second specialist at GSTT where diagnosis was uncertain. The videos captured in real clinical environments, resulting in variability in camera equipment, camera placement, recording conditions and background activity. Prior to

processing each video was temporally clipped to remove frames with nursing staff or carers or movement after the task was performed.

To enable computational analysis and ensure data privacy, videos were processed using a pose estimation pipeline (OpenPose) to extract two-dimensional skeletal keypoints for 25 anatomical joints [17]. This converts raw video into structured sequences describing body posture and motion over time while preserving key spatial and temporal relationships. Sequences were segmented into fixed-length temporal windows as inputs to the spatio-temporal model, resulting in varying numbers of samples per patient depending on recording length.

2.2.2 Data Processing and Pipeline Correction

Data Leakage Prevention

In the original training pipeline, the dataset was divided into fixed-length sequences before the train-test split. This allowed sequences from the same patient to appear in both sets, introducing data leakage due to shared temporal continuity and motion patterns.

To address this, the train-test split was performed at the patient level prior to sequence division, ensuring that entire recordings were assigned to either the training or testing set. Sequence division was then applied independently within each subset, preventing overlap between sets. This ensures evaluation on unseen patient recordings and proves a more realistic estimate of generalisation performance.

Frame Rate Processing

Temporal resolution is important in modelling movement patterns. In the original implementation, frame subsampling introduced uneven temporal gaps within sequences, resulting in clusters of consecutive frames being retained while others were removed. This produced non-uniform temporal sampling and disrupted the continuity of motion patterns.

To address this, a revised frame sampling strategy was implemented to ensure temporal coverage that was as uniform as achievable without interpolation between frames. Notes theoretically there are some subsampling paradigms where achieving uniform frame rate is not possible, and therefore for those the algorithm, takes the nearest possible frame rate rather than removing frames in large sections, frames are retained at regular intervals across the sequence. Figure 3 compares the original implementation with the present implementation.

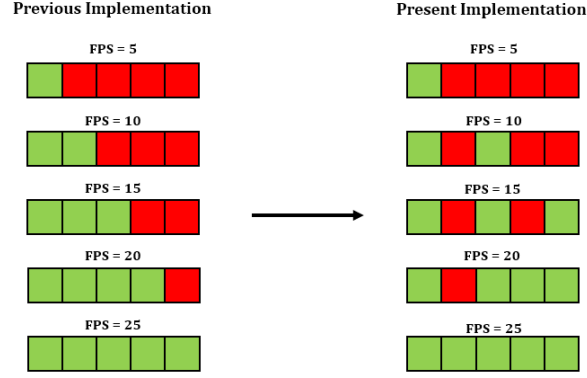


Figure 3: Previous frame rate calculations vs present frame rate calculations for a 5-frame video clip

This approach ensures that temporal information is sampled more consistently across the recording, allowing the temporal network to observe motion dynamics with similar time between frames over the full duration of the sequence rather than from isolated clusters of frames with variable speed.

Spatial Normalisation

Clinical recordings were collected using two camera configurations with different aspect ratios, leading to variation in pose-estimated coordinates. Without correction, the model could learn camera-dependent features rather than clinically meaningful movement patterns.

To address this, spatial normalisation was applied to skeletal keypoints before model input. Each skeleton was translated so that the “MidHip” joint was centred at the origin, removing global positional variation.

The skeleton was then scaled by shoulder width (distance between left and right shoulders) to standardise body size while preserving joint geometry. Finally, orientation was adjusted so the shoulder line was horizontal, reducing variation due to camera tilt or subject orientation.

These transformations ensure the model learns movement dynamics relative to the body rather than camera-dependent artefacts.

Data Augmentation

To improve generalisation and mitigate limited dataset size, data augmentation was applied to the training data. Augmentation was paired with normalisation to minimise residual features related to aspect ratio and ensure consistency across different camera setups. Augmentation was performed after the train-test split to prevent data leakage.

Three augmentation strategies were implemented – one in the baseline and two additional methods as preprocessing experiments. First, horizontal flipping of skeletal keypoints was applied to simulate mirrored body movements while preserving anatomical relationships. This transformation increases dataset variability while maintaining clinically meaningful movement patterns.

Second, Gaussian noise augmentation was implemented by adding small random perturbations to visible keypoint coordinates. This simulates minor variations introduced by pose estimation inaccuracies and improves model robustness to noise in keypoint detection.

Third, the joint dropout augmentation was introduced to simulate temporary occlusion or missing keypoints. This was achieved by randomly removing a small proportion of visible joints during training, encouraging the model to learn robust movement representations from incomplete skeletal data.

Each augmentation strategy was applied independently during ablation experiments to evaluate their individual contributions to model performance. Augmentation was applied only to training data, and original samples were retained alongside augmented versions, effectively increasing dataset size while preserving the original distribution.

2.3 Spatio-Temporal Model

2.3.1 Network Architecture

The model used in this study is a pose-based spatio-temporal neural network designed to classify hyperkinetic movement disorders from skeletal keypoint sequences extracted from clinical video recordings. The architecture follows the framework proposed in previous work by Nandika et al. [15], combining spatial modelling of body joint relationships with temporal modelling of movement dynamics.

The spatial components (top pathway in Figure 4) represent the human skeleton as a graph, where nodes correspond to anatomical keypoints and edges represent biomechanical connections between joints. A graph neural network processes these skeletal structures to learn spatial relationships between body joints within each frame sequence.

Temporal dynamics are modelled using a bidirectional Long Short-Term Memory (BiLSTM) network (bottom pathway in Figure 4), which captures motion patterns across time [41]. A temporal self-attention mechanism is applied to the LSTM outputs to allow the model to assign different importance to individual timesteps within the sequence [25]. This enables the network to focus on movement segments that are most informative for classification.

The outputs of the spatial and temporal networks are concatenated (right pathway in Figure 4) and passed to a fully connected layer, batch normalisation and leaky ReLU, followed by a final fully connected layer and softmax which is threshold at 0.6 to produce the final prediction for dystonia-predominant or chorea-predominant labels.

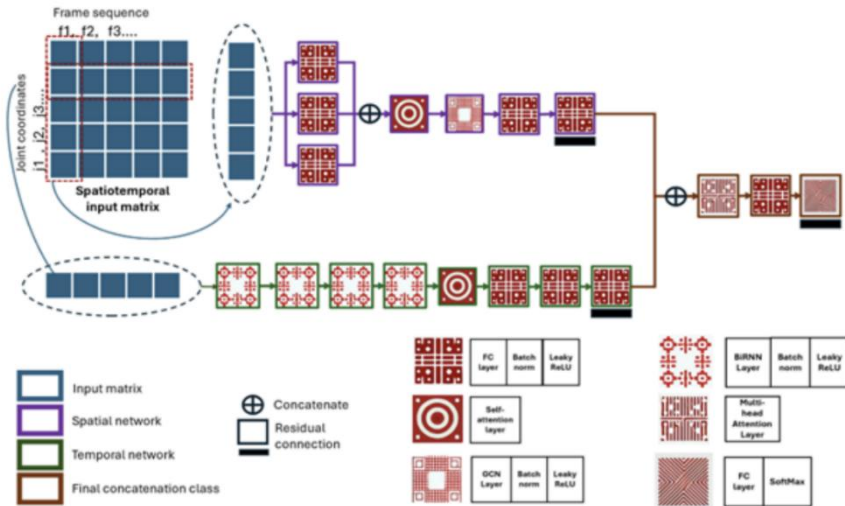


Figure 4: A block diagram of the network architecture. FC Layer = Fully Connected Layer, BiRNN = Bidirectional RNN

2.3.2 Training Procedure

The network was trained using supervised learning with the Adam optimiser and weight decay and dropout as forms of regularisation [42]. A cyclic learning rate was used during training, periodically increasing and decreasing the learning rate throughout optimisation. This approach helps the optimiser escape suboptimal local minima and plateaus in the solution space by temporarily increasing the step size, allowing the model to explore alternative regions of the parameter space before gradually reducing the learning rate again.

To address class imbalance within the dataset, focal loss was used as the training objective. Focal loss modifies cross-entropy loss by introducing a focussing parameter (γ), which emphasise difficult training examples [43]. While the original architecture incorporated the focusing term, the present study extends the formulation by introducing the class weighting parameter α . This allows different classes to contribute unequally to the loss function.

The minority class (chorea-predominant) was assigned a higher weight, while the majority class (dystonia) received a lower weight. This encourages the model to prioritise minority class errors during optimisation, to reduce the effect of the imbalanced dataset.

2.4 Temporal Attention and Interpretability

2.4.1 Temporal Attention Behaviour

The temporal component of the model incorporated a self-attention mechanism applied to the outputs of the bidirectional LSTM. This mechanism assigns a learnable importance weight to each timestep in the input sequence, allowing the network to emphasise frames that contribute most strongly to the classification decision.

For a sequence of hidden states produced by the LSTM, the attention module computes a set of attention scores that are normalised using a softmax function to produce attention weights across the temporal dimension. These weights are then used to compute a weighted representation of the sequence, allowing the model to focus on informative segments of motion while suppressing less relevant frames.

Understanding how the model allocates attention across time is particularly important in the context of HMDs, where diagnostically relevant motion patterns may occur intermittently or continuously throughout the recording depending on the studied disease. Analysing the temporal attention behaviour therefore provides insight into whether the model focuses on clinically meaningful movement events or on false temporal features.

To analyse this behaviour, the attention weights generated during inference were recorded for each sequence. These weights were visualised as temporal heatmaps, enabling the analysis of how attention is distributed across frames within each movement sequence.

2.4.2 Interpretability Techniques

To further investigate the temporal reasoning of the model, three complementary interpretability techniques were applied: temporal attention maps, saliency maps and Gradient-weighted Class Activation Mapping (Grad-CAM).

Attention maps provide a direct visualisation of the attention weights produced by the model's self-attention module. These maps indicate how strongly the model attends to each timestep when forming its internal representation of the sequence. However, attention alone does not reflect the direct contribution of each frame to the final classification output [25].

To complement this, saliency maps were computed by calculating the gradient of the model prediction with respect to the input features. This gradient measures how sensitive the predicted class is to small changes in the input sequence, highlighting which timesteps most strongly influence the model's internal representation [44].

Grad-CAM was also applied to the temporal network to identify which model weights contributed most strongly to the final classification decision. Unlike saliency maps, which measure model sensitivity to the input, Grad-CAM highlights regions of the intermediate feature representations that are most important for predicting a given class [33].

By comparing these three visualisation techniques, it is possible to analyse how the model processes temporal information and determine whether attention-based explanations align with gradient-based methods leading to greater insights into model behaviour. Comparing these methods provides insight into how the model processes temporal information and allows discrepancies between attention patterns and true predictive importance to be identified.

2.5 Ablation Study Design

To evaluate the contribution of individual components within the spatio-temporal architecture, a series of ablation experiments were conducted. Ablation studies systematically remove or modify elements of a model to determine their influence on overall performance.

Two primary categories of ablation were explored: architectural ablations and preprocessing ablations.

The architectural ablation study examines the role of the spatial and temporal components of the network. In these experiments, either the spatial graph neural network or the temporal BiLSTM module was disabled while the remaining components were retained. This allows the contribution of the spatial joint relationships and temporal motion patterns to the final prediction to be assessed independently.

In addition to removal-based ablations, the temporal modelling component was further investigated by replacing the BiLSTM with alternative sequential architectures. Specifically, the BiLSTM was substituted with a Gated Recurrent Unit (GRU) in one configuration and a transformer-based temporal model in another. A GRU is a type of recurrent neural network that uses gating mechanisms to regulate flow of information through time, enabling efficient modelling of temporal dependencies while reducing the impact of vanishing gradients. In contrast, a transformer is a neural network architecture that uses self-attention mechanisms to model relationships across an entire sequence without relying on recurrent architectures, allowing it to capture long-range temporal dependencies more effectively. These substitutions allow for comparison between different approaches to temporal feature learning, assessing whether model performance is dependent on the specific recurrent structure or can be improved through alternative architectures.

The preprocessing experiments were used to evaluate the effect of different input processing strategies. These experiments investigated how modifications to frame sampling, spatial normalisation (to evaluate how model robustness is affected by reducing camera-dependent spatial variations) and data

augmentation (to investigate whether artificially increasing dataset variability improves generalisation, affect model performance).

By comparing performance across these configurations, the ablation studies provide insight into which components contribute most strongly to classification accuracy and help ensure model generalisation.

2.6 Evaluation Strategy

Model performance was evaluated using an unseen test set to assess generalisation. During development, stratified k-fold cross-validation was used to validate model stability and guide parameter selection. A fixed train-test split was then used for all subsequent experiments, ensuring consistent comparison across model configurations and preprocessing strategies.

Multiple metrics were used to evaluate classification performance. While overall accuracy was reported, it may be misleading due to class imbalance. Therefore, balanced accuracy was used as the primary metric, as it accounts for both sensitivity and specificity.

Additional metrics included sensitivity and specificity, reflecting performance on dystonia- and chorea-predominant cases, respectively, along with F1-score to capture the balance between false positives and false negatives. ROC-AUC was also reported as a threshold-independent measure of class separability.

Together, these metrics provide a balanced evaluation of model performance and allow meaningful comparison between architectural modifications and preprocessing strategies explored in the ablation experiments.

3 Experimental Results

To ensure consistent comparison across experiments, each model configuration is evaluated using the same set of performance metrics. For each ablation, overall accuracy, sensitivity, specificity, F1-score and ROC-AUC are reported. Confusion matrices are also presented where appropriate.

3.1 Baseline Model Performance

The baseline spatio-temporal model was evaluated on an unseen test dataset. Performance metrics are summarised in Table 1.

Model	Sensitivity	Specificity	Balanced Accuracy	F1-Score	ROC-AUC
Baseline	0.893	0.824	0.858	0.915	0.898

Table 1: Baseline model performance statistics. Results shown for a single fixed train-test split.

Figure 5 shows the corresponding confusion matrix for the baseline model. The model achieved correct classification for 92 out of 103 dystonia-predominant samples and 28 out of 34 chorea-predominant samples.

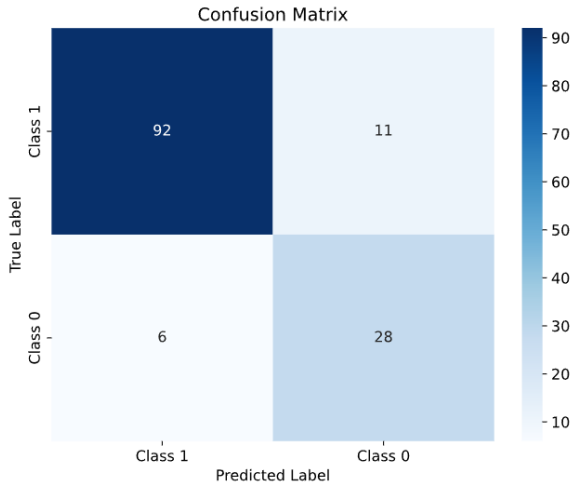


Figure 6: Confusion matrix of baseline results (Class 1 = Dystonia, Class 0 = Chorea)

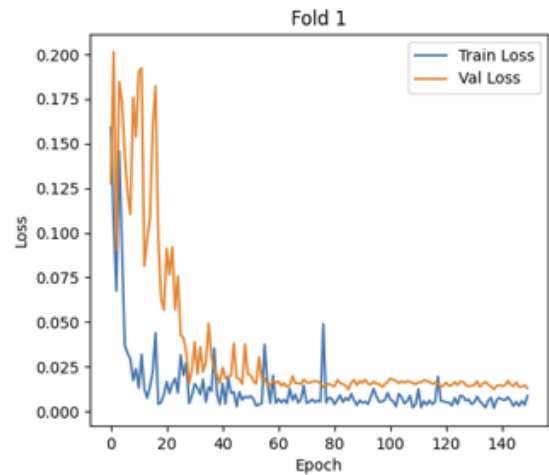


Figure 5: Training and validation loss plots (Baseline)

The training and validation loss curves are presented in Figure 6, showing convergence after approximately 60 epochs with a small gap between training and validation loss. Early instability in the loss curves is attributed to the higher learning rates introduced by the cyclic learning rate schedule (triangular2 mode).

Temporal Grad-CAM, temporal attention and saliency visualisation were generated for interpretability analysis. Representative examples are shown in Figures 7-9. Note the heatmaps scale differently between each figure with the Temporal Attention (Fig 8) having the smallest range and the Saliency Map (Fig 9) having the largest range.

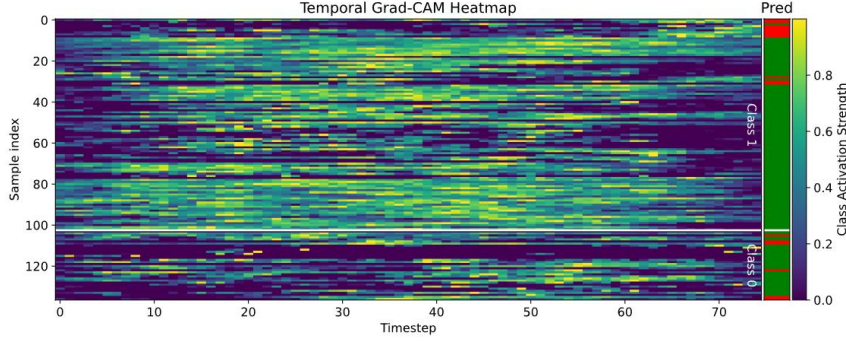


Figure 7: Temporal Grad-CAM heatmap (Baseline) showing activation. Higher activation is represented by warmer colours. The central “Pred” bar indicates classification outcome (green = correct, red = incorrect).

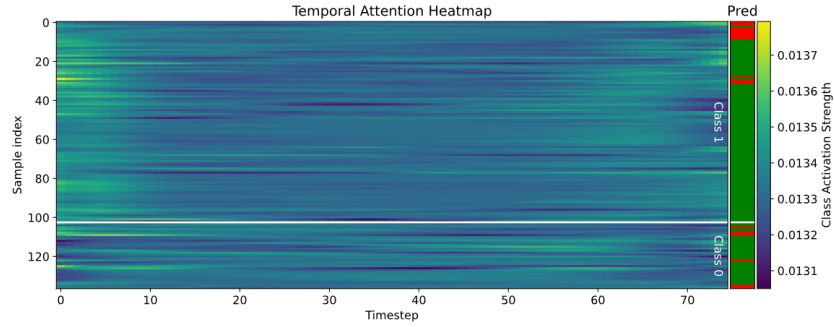


Figure 8: Temporal attention heatmap (Baseline). Higher attention is represented by warmer colours. The central “Pred” bar indicates classification outcome (green = correct, red = incorrect).

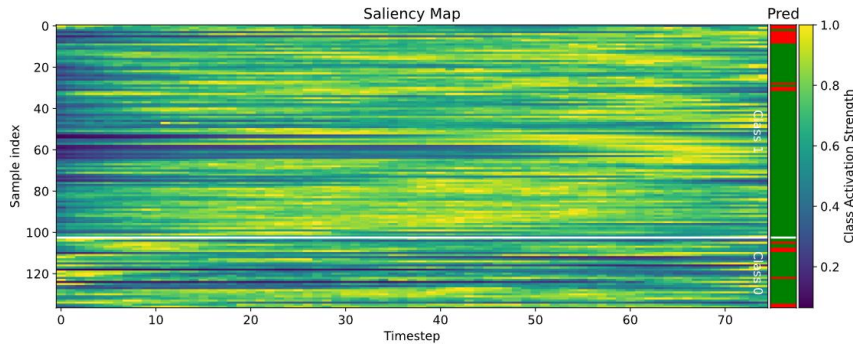


Figure 9: Saliency heatmap (Baseline). Higher saliency values are represented by warmer colours. The central “Pred” bar indicates classification outcome (green = correct, red = incorrect).

In Figure 7 (Temporal Grad-CAM Heatmap), higher activation corresponds to brighter colours (yellow/green), while lower activation is represented by darker colours (blue). The y-axis represents individual samples, ordered such that dystonia-predominant sequences occupy the upper portion of the plot and chorea-predominant sequences occupy the lower portion, with the two classes separated

by a white line. Within the dystonia samples (upper rows), regions of high activation appear more sustained across time, forming broader, continuous bands. In contrast, chorea-predominant samples (lower rows) exhibit more intermittent activation, characterised by shorter, fragmented peaks of high intensity separated by lower activation regions.

This distinction is less clearly observed in Figure 8 (Temporal Attention Heatmap) and Figure 9 (Saliency Heatmap), where activation appears more diffuse and less temporally structured across both classes. This suggests that the Grad-CAM representation provides a clearer indication of temporally discriminative features learned by the model compared to attention and saliency-based methods.

3.2 Architectural Ablation Results

To evaluate the contribution of individual components, a series of ablation experiments were performed on the baseline spatio-temporal network. These experiments examined the effect of removing the spatial module, removing the temporal module, removing the temporal attention and replacing the bidirectional LSTM with alternative temporal architectures. Results were evaluated on the same fixed train-test split to ensure fair comparison across configurations. Performance for all architectural variants is summarised in Table 2, with the proposed baseline model giving the best overall balanced accuracy. No temporal module gave the best specificity, i.e. ability to recognise chorea-predominant samples. The best sensitivity was achieved with using a unidirectional LSTM.

Model	Sensitivity	Specificity	Balanced Accuracy	F1-Score	ROC-AUC
Baseline	0.893	0.824	0.859	0.915	0.898
No Spatial Module	0.816	0.588	0.702	0.836	0.851
No Temporal Module	0.631	0.912	0.772	0.760	0.813
No Temporal Attention	0.602	0.735	0.669	0.713	0.717
Unidirectional LSTM	0.971	0.529	0.750	0.913	0.823
GRU	0.796	0.824	0.810	0.859	0.795
Transformer	0.757	0.794	0.776	0.830	0.896

Table 2: Architectural ablation performance statistics. Results shown for a single fixed train-test split.

To further investigate how architectural modifications influenced model behaviour, interpretability heatmaps were generated for selected experiments. Representative temporal attention/saliency and Grad-CAM visualisation are shown in Figures 10 and 11.

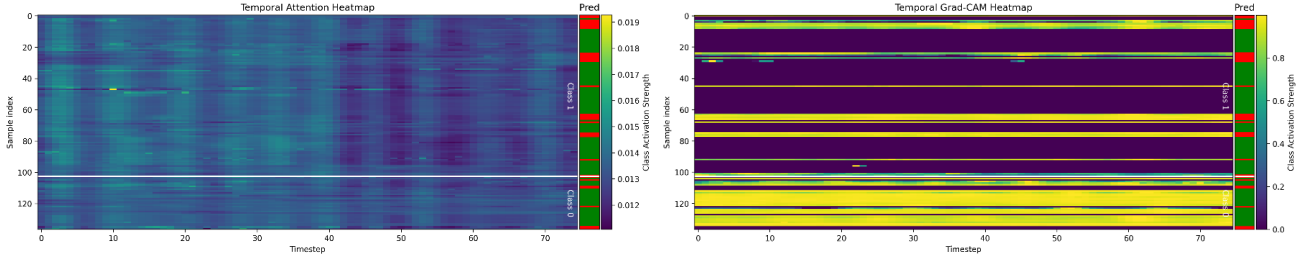


Figure 10: Saliency heatmap (left) and Grad-CAM heatmap (right) for the described transformer architecture ablation.

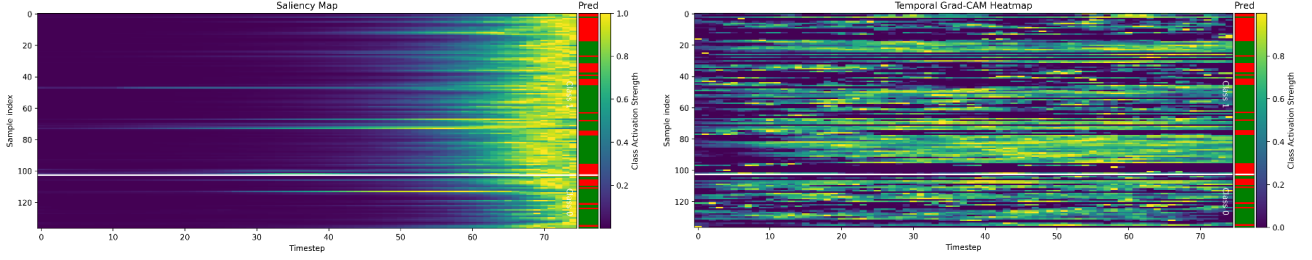


Figure 11: Saliency heatmap (left) and Grad-CAM heatmap (right) for the described temporal attention ablation.

The temporal attention ablation model demonstrates more diffuse activation patterns, whereas the model with a transformer module in place of the BiLSTM exhibits broader temporal activation across the sequence compared to the baseline module.

3.3 Pre-processing Study Results

To further evaluate the impact of preprocessing techniques on model performance, additional experiments were conducted varying how the data were processed prior to model input. These experiments assessed the influence of spatial normalisation, data augmentation and frame rate selection on model performance. For each configuration, performance metrics were computed using the same fixed train-test split to ensure fair comparison.

Normalisation

To evaluate the contribution of spatial normalisation, the model was trained without applying the normalisation procedure described in Section 2.2.2. All other training parameters and architectural components were kept identical to the baseline configuration.

Performance metrics for the model trained without spatial normalisation are reported in Table 3. This experiment assesses whether normalising skeletal keypoints improves model robustness by reducing camera-dependent spatial variation. No data normalisation significantly reduced model performance on all performance metrics.

Model	Sensitivity	Specificity	Balanced Accuracy	F1-Score	ROC-AUC
No Normalisation	0.621	0.706	0.664	0.723	0.650

Table 3: Described normalisation experiments performance statistics. Results shown for a single fixed train-test split.

Augmentation

To evaluate the effect of data augmentation on model performance, multiple augmentation strategies were evaluated, as described in Section 2.2.2. This includes no augmentation, flip (baseline), noise and flipping, noise, flipping and joint dropout to determine the effect of the applied augmentation in the baseline.

Performance metrics for each augmentation configuration are reported in Table 4. These experiments assess whether artificially increasing dataset variability improves generalisation performance. The baseline approach achieved the best performance, however, adding noise & dropout only slightly degraded the model performance in terms of classifying chorea-predominant and dystonia-predominant cases. No data augmentation performed significantly worse across all performance metrics.

Data Augmentation	Sensitivity	Specificity	Balanced Accuracy	F1-Score	ROC-AUC
No Augmentation	0.825	0.500	0.663	0.829	0.791
Noise + Flip	0.893	0.765	0.829	0.906	0.863
Noise + Flip + Dropout	0.893	0.765	0.829	0.906	0.894
Baseline	0.893	0.824	0.858	0.915	0.898

Table 4: Described augmentation experiments performance statistics. Results shown for a single fixed train-test split.

Frame Rate

To evaluate the impact of temporal resolution on classification performance, additional experiments were conducted using different frame sampling rates. The baseline model used a frame rate of 25 frames per second.

Four additional frame rates were evaluated at 5, 10, 15 and 20 frames per second, with all other model parameters held constant. This allows assessment of how temporal resolution influences the model’s ability to capture movement dynamics.

Performance metrics for each frame rate configuration are reported in Table 5. These experiments investigate whether increasing temporal resolution improves classification performance or introduces additional noise. Whilst the 5 FPS model achieved the best performance across most metrics, the baseline model shows the most consistent results, whilst also utilising all input data – many samples were not included during the 5 FPS experiment therefore does not show reliable results.

Model	Sensitivity	Specificity	Balanced Accuracy	F1-Score	ROC-AUC
5 FPS	1.000	0.800	0.900	0.970	0.863
10 FPS	0.919	0.643	0.781	0.895	0.865
15 FPS	0.915	0.700	0.808	0.908	0.864
20 FPS	0.841	0.690	0.766	0.862	0.856
25 FPS (Baseline)	0.893	0.824	0.858	0.915	0.898

Table 5: Described frame rate experiments performance statistics. Results shown for a single fixed train-test split.

4 Discussion

4.1 Overview of Key Findings

This study evaluated a spatio-temporal deep learning architecture for classification of dystonia-predominant and chorea-predominant movement disorders using skeletal keypoints extracted from a widely used pose estimation model. Balanced accuracy was prioritised in the following discussion as a performance metric due to class imbalance, ensuring that performance reflected both dystonia and chorea classifications. ROC-AUC values were consistent with balanced accuracy trends, supporting the robustness of model discrimination between dystonia and chorea.

The baseline model achieved the highest overall performance, with a balanced accuracy of 85.9%, demonstrating effective classification between the two phenotypes. The confusion matrix indicated strong classification performance for both classes, with particularly high sensitivity for dystonia-predominant cases and robust specificity for chorea-predominant cases, suggesting that the model captured distinguishing movement characteristics between the two disorders.

Architectural ablation experiments demonstrated that both the spatial and temporal components contributed meaningfully to classification performance. Removal of the spatial module resulted in a substantial decrease in balanced accuracy indicating that joint-relationships and body-wide coordination patterns are important for distinguishing movement phenotypes. Similarly, removal of the temporal module reduced model performance highlighting the importance of temporal movement dynamics in differentiating dystonia and chorea. Removal of temporal attention produced one of the largest performance reductions, suggesting that attention mechanisms improved the model’s ability to identify informative temporal segments within movement sequences.

Alternative temporal architectures produced varied performance outcomes. The use of a GRU for the temporal module in the model achieved performance closest to the baseline, indicating that simpler recurrent architectures may still capture some meaningful temporal dependencies, with a slight degradation in performance. In contrast, the unidirectional LSTM demonstrated reduced specificity, suggesting that bidirectional temporal context improves classification stability. The model with a Transformer module demonstrated moderate performance, with heatmap visualisation suggesting broader temporal activation patterns, consistent with more global temporal representations.

Preprocessing ablation experiments further demonstrated the importance of input preparation in model performance. Data augmentation experiments improved performance compared to training without augmentation, indicating that artificially increasing dataset variability improved model

generalisation. The application of additional augmentation strategies demonstrated similar performance, suggesting diminishing returns from increasingly complex augmentation pipelines.

Frame rate ablation experiments demonstrated variable performance across temporal sampling rates. Although the lowest frame rate achieved the highest balanced accuracy, no consistent trend was observed across frame rate configurations. Given the relatively small dataset, which was reduced even more at lower frame rates, these results should be interpreted cautiously, as performance differences may reflect variability in training and testing samples rather than meaningful improvements in model performance at higher temporal resolutions.

4.2 Clinical Interpretation of Model Behaviour

Interpretability analysis of the baseline model provides insight into how the network differentiates between dystonia-predominant and chorea-predominant movement disorders. Temporal attention and Grad-CAM visualisations demonstrated distinct activation patterns between classes, suggesting that the model learned clinically meaningful movement dynamics rather than relying solely on static posture information.

Dystonia-predominant sequences typically demonstrated broader and more sustained activation across the temporal sequence. This behaviour is consistent with the clinical presentation of dystonia, which is characterised by sustained, abnormal postures, co-contraction of opposing muscle groups and gradual evolution of movement patterns. These movements often persist across multiple frames, resulting in continuous activation patterns and suggesting that the model identifies prolonged deviations in joint positioning and coordination as key discriminative features.

In contrast, chorea-predominant sequences demonstrated more fragmented and intermittent activation patterns. Chorea is characterised by brief, irregular and unpredictable movements that occur intermittently across different body regions. The model therefore appeared to focus on shorter temporal segments and rapidly changing joint dynamics, consistent with the transient nature of choreiform movements.

The spatial and temporal components of the model likely contribute jointly to these distinctions. The spatial module captures coordination across joints, while the temporal module captures the evolution of movement patterns over time. This is supported by the ablation results, with reduced sensitivity in the spatial ablation model and reduced specificity in the temporal ablation model.

Interpretability analysis of architectural ablations further supports this interpretation. Removal of temporal attention resulted in more diffuse activation patterns, suggesting reduced ability to identify clinically relevant movement segments. Similarly, the model with a Transformer module demonstrated

broader activation across sequences, suggesting a more global representation of movement patterns that may reduce its ability to short duration choreiform movements.

The comparison of temporal interpretation techniques reveals that while the model consistently identifies clinically relevant features, the expressive power of these visualisation methods differs. Grad-CAM produced sharp, interpretable mappings that correspond closely to the physical manifestations of dystonia-predominant and chorea-predominant cases. However, the temporal attention mechanisms yielded a more attenuated signal, with subtle weight shifts that lack the definitive contrast seen in Grad-CAM and some Saliency Maps. This suggests that while the model's internal logic is biologically grounded, Grad-CAM may be the more effective interpretation technique for extracting actionable clinical insights from the underlying architecture.

4.3 Ablation Study Interpretation

Architectural ablation experiments demonstrated that both spatial and temporal modelling components contributed meaningfully to classification performance. Removal of the spatial module reduced balanced accuracy from 85.9% in the baseline model to 70.2%, representing one of the largest performance decreases. This suggests that joint relationships and body-wide coordination patterns are important for distinguishing movement phenotypes, which is consistent with clinical observations.

Removal of the temporal module also reduced performance, with balanced accuracy decreasing to 77.2%. While specificity increased to 91.2%, sensitivity dropped substantially to 63.1%, suggesting that the model became more conservative in identifying dystonia-predominant cases. This finding highlights the importance of temporal movement dynamics in differentiating between dystonia and chorea, as these disorders are defined not only by posture but also by how movement evolves over time.

Removal of the temporal attention produced decreased balanced accuracy to 66.9%. Sensitivity also fell to 60.2%, indicating reduced ability to correctly classify dystonia-predominant cases. This was accompanied by more diffuse activation patterns in interpretability visualisations, suggesting that temporal attention improves the model's ability to identify informative movement segments, such as sustained dystonic postures or intermittent choreiform bursts.

Alternative temporal architectures demonstrated varied performance outcomes. The GRU in the temporal module achieved a balanced accuracy of 81.0%, performing closest to the baseline model using a bidirectional LSTM and suggesting that simpler recurrent architectures may still capture meaningful temporal dependencies. In contrast, the unidirectional LSTM demonstrated reduced specificity (52.9%) despite high sensitivity (97.1%), resulting in a lower balanced accuracy of 75%. This suggests that bidirectional temporal context improves classification stability by incorporating both preceding and subsequent movement information.

Heatmap visualisations demonstrated broader and more uniform temporal activation patterns when using a Transformer, with Grad-CAM maps appearing almost binary across correctly classified sequences. This suggests that the Transformer may rely on a more global temporal interpretation, identifying overall movement characteristics rather than focussing on specific informative segments. In some cases, this behaviour may indicate that the model determines class membership early within the sequence and maintains this decision across subsequent frames, resulting in uniformly high activation throughout the clip.

While this global representation may capture overall movement patterns, it may reduce sensitivity to short-duration movement events such as intermittent choreiform bursts or evolving dystonic postures. Additionally given the relatively small dataset and short clip durations, the Transformer may be limited in learning fine-grained temporal dependencies, as these architectures typically benefit from larger datasets [28]. These factors may explain the slightly reduced performance compared to the bidirectional LSTM temporal module used in the baseline model.

These findings indicate that accurate classification of hyperkinetic movement disorders relies on both spatial coordination patterns and temporal movement dynamics with attention mechanisms further improving identification of clinically relevant movement segments.

4.4 Pre-processing Study Interpretation

Preprocessing experiments further highlighted the importance of input preparation in model performance. Removal of spatial normalisation resulted in a substantial decrease in balanced accuracy from 85.9% to 66.4%, indicating that normalising skeletal keypoints improves robustness and reduces camera-dependent spatial variation when training the model. Without normalisation, the model may inadvertently learn recording-specific spatial characteristics rather than clinically meaningful movement patterns, reducing generalisation performance.

Data augmentation experiments demonstrated improved performance compared to training without augmentation. Removal of augmentation reduced balanced accuracy to 66.3%, while the application of noise and horizontal flip augmentations increased balanced accuracy to 82.9%. These findings suggest that augmentation improves model robustness to variations in patient positioning, pose estimation noise and recording conditions. The addition of dropout for specific skeleton keypoints did not further improve performance, indicating diminishing returns from increasingly complex augmentation strategies.

Frame rate ablation experiments demonstrated variable performance across temporal sampling rates. The 5 FPS configuration achieved the highest balanced accuracy of 90.0%, while 10, 15 and 20 FPS achieved 78.1%, 80.8% and 76.6% respectively. However, no consistent trend was observed across

frame rates. Given the relatively small dataset and evaluation using a single train-test split, these findings should be interpreted cautiously, as performance differences may reflect sampling variability rather than meaningful improvements in temporal resolution.

These findings suggest that preprocessing choices play an important role in model performance, with spatial normalisation and augmentation providing consistent improvements, while optimal temporal resolution remains uncertain given dataset limitations.

4.5 Limitations and Dataset Bias

Several limitations should be considered when interpreting these results. The dataset size was small, which may limit generalisability. Although the baseline model achieved strong performance, evaluations were conducted using a single train-test split, meaning results may be sensitive to sample distribution. This is particularly relevant for the frame rate experiments, where variability may reflect sampling differences rather than true performance changes.

The dataset was also imbalanced, with more dystonia-predominant samples than chorea-predominant samples. This may contribute to higher sensitivity for dystonia and reduced specificity for chorea in some experiments. Although balanced accuracy and focal loss were used, class imbalance may still influence model behaviour. Additionally, varying video durations results in unequal sample contributions per patient, which may introduce bias.

The dataset was also derived from clinical recordings collected at a single specialist centre, which may introduce referral bias. Additionally, socioeconomic and demographic factors influence referral patterns, potentially limiting dataset diversity. For example, children from certain ethnic or socioeconomic backgrounds are less likely to be referred for specialist evaluation and have greater challenges attending specialist clinics due to travel and caring expenses, which may reduce representation of specific cohorts within the dataset. These factors may affect model generalisation when applied to broader populations. Finally, the study used binary classification, whereas movement disorders often exist on a spectrum with overlapping phenotypes.

4.6 Clinical Implications and Future Work

The findings of this study demonstrate the potential of spatio-temporal deep learning models for classifying hyperkinetic movement disorders using pose-based skeletal data. By capturing clinically meaningful spatial and temporal features, this approach may support more objective assessment of dystonia and chorea, which are traditionally reliant on subjective clinical interpretation. Automated classification tools may therefore improve diagnostic consistency, particularly in settings with limited

specialist expertise. Interpretability techniques further support clinical adoption by providing transparent visual explanations of model predictions.

Pose-based representations also offer practical advantages for clinical deployment, enabling analysis of standard video recordings without specialised sensors. This supports retrospective analysis of existing data and potential integration into telemedicine workflows, with applications in screening, monitoring disease progression and evaluating treatment response.

Future work should focus on expanding dataset size and diversity through multi-centre data collection, alongside exploration of longer temporal sequences to better capture movement dynamics. Hybrid architectures combining recurrent and attention-based approaches may further improve temporal modelling by capturing both short- and long-term dependencies.

Extending beyond binary classification to include mixed phenotypes would better reflect clinical practice. Prospective clinical validation and investigation of real-time implementation will be important steps towards clinical translation.

5 Conclusion

This study evaluated and optimised a spatio-temporal neural network for classifying paediatric hyperkinetic movement disorders, addressing limitations in interpretability, robustness and model evaluation. A pose-based pipeline was implemented and analysed using ablation studies, preprocessing experiments and interpretability techniques to assess the contribution of spatial and temporal components.

The baseline model achieved strong performance, with a balanced accuracy of 85.9%. Ablation results showed that both spatial and temporal components are essential, with performance decreasing when either was removed. Temporal attention mechanisms further improved performance, highlighting the importance of identifying informative movement segments.

Limitations include small dataset size and short temporal sequences, which restrict the modelling of long-term dynamics. Future work should focus on larger, multi-centre datasets, improved temporal modelling and extension beyond binary classification, alongside clinical validation deployment.

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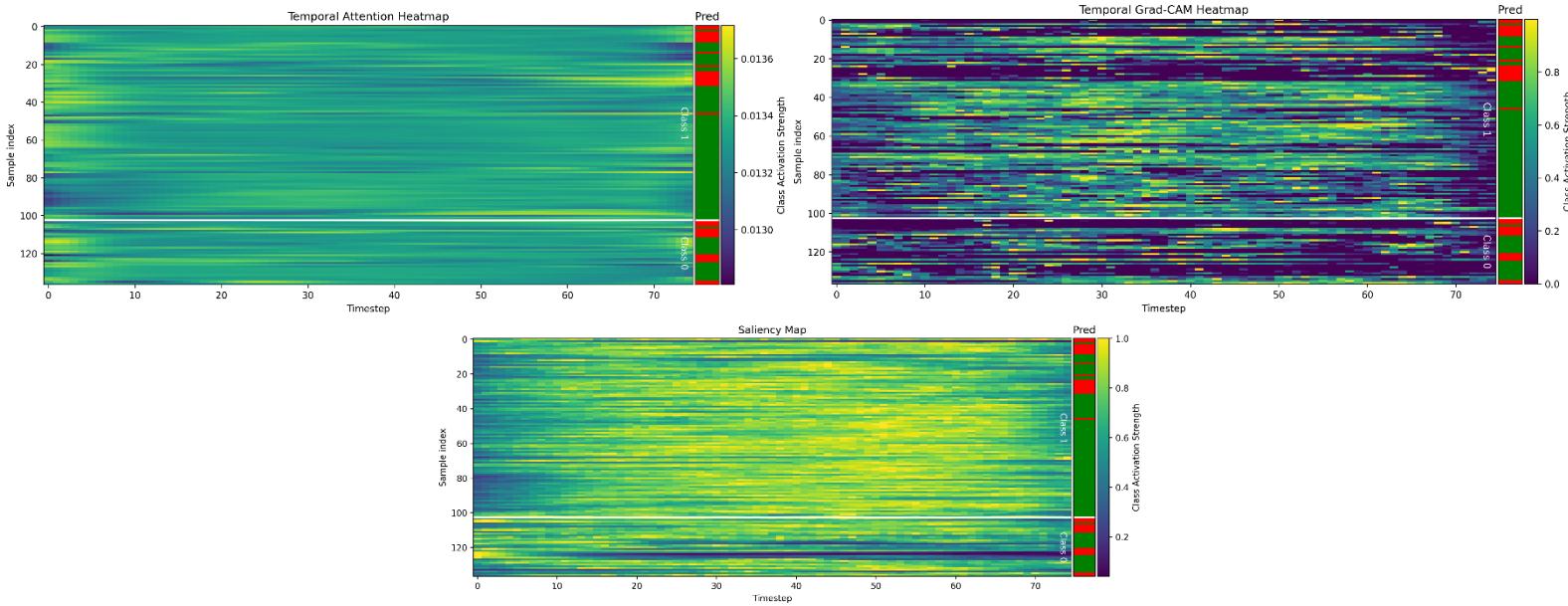
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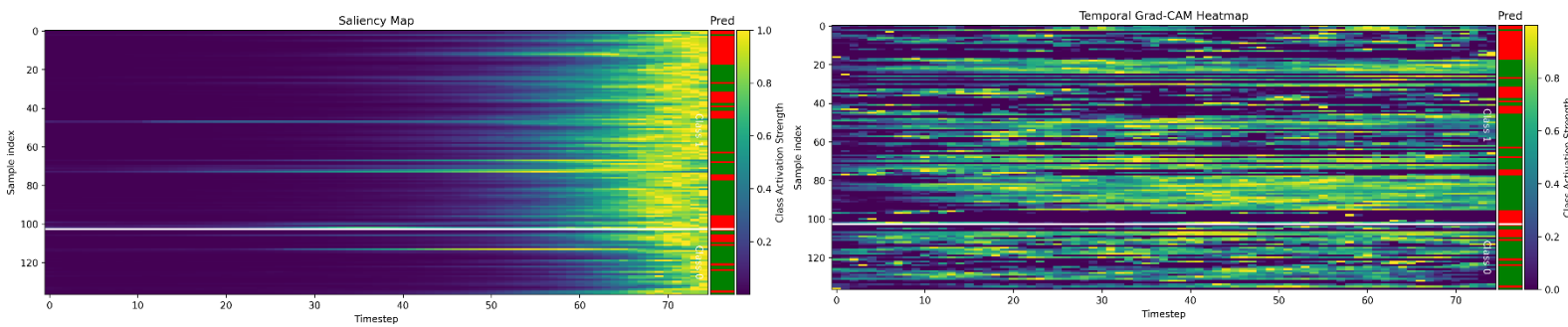
7 Appendices

7.1 Appendix A

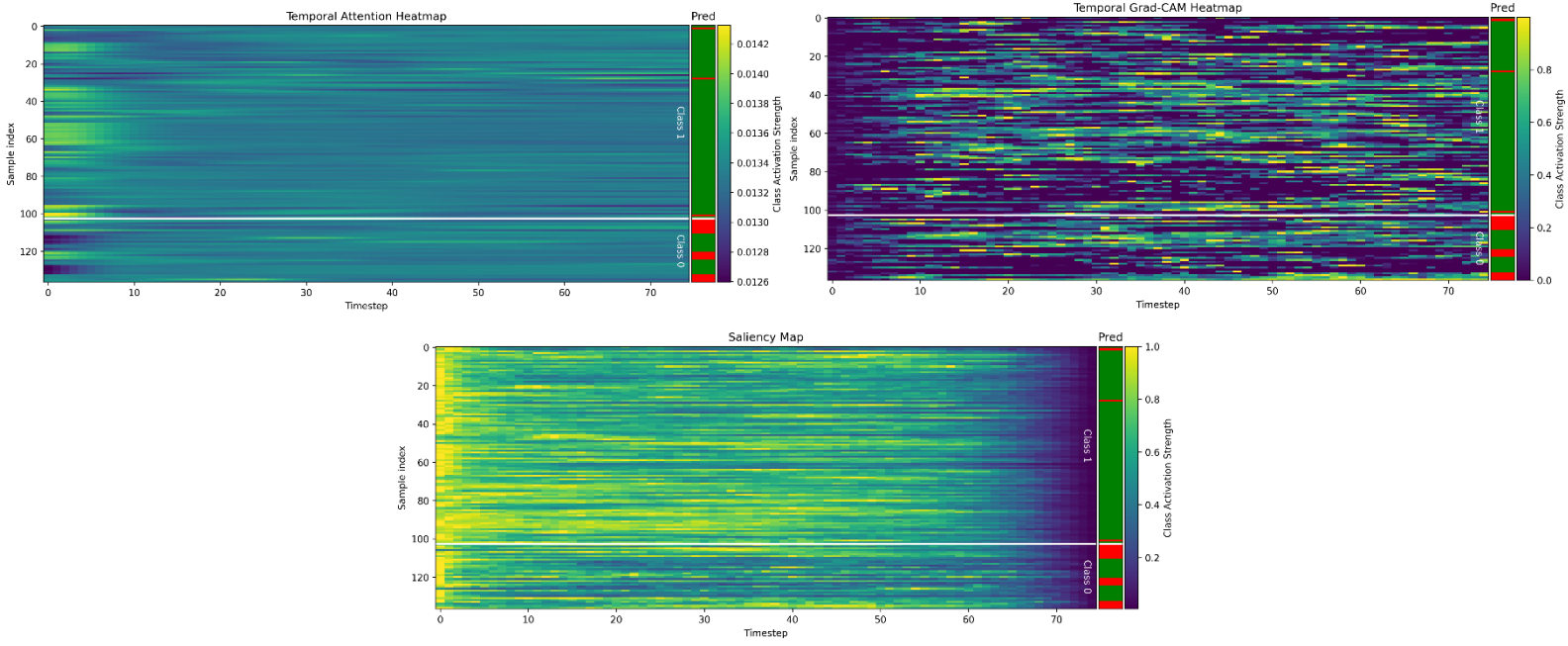
Supplementary Figure 1: Saliency and Grad-CAM Heatmaps for described Spatial Ablation



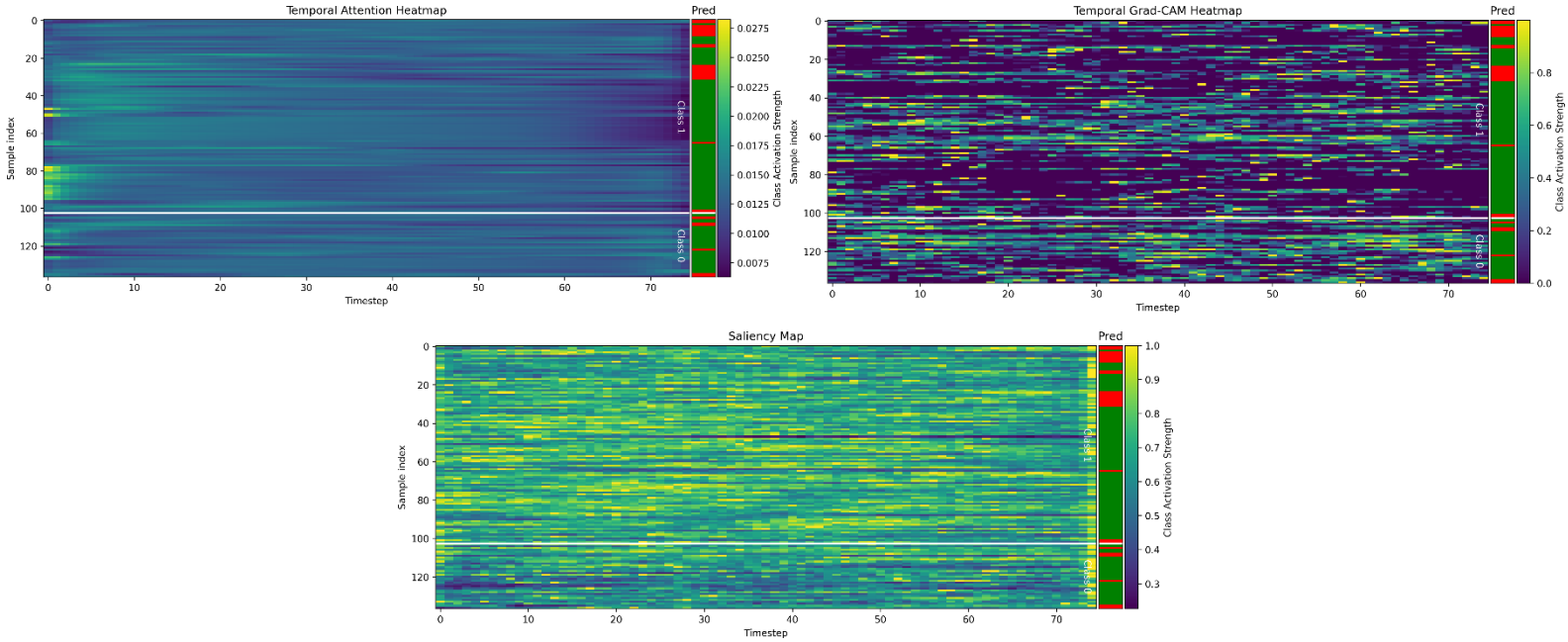
Supplementary Figure 2: Saliency and Grad-CAM Heatmaps for described Temporal Attention Ablation



Supplementary Figure 3: Attention, Saliency and Grad-CAM Heatmaps for described Unidirectional Ablation



Supplementary Figure 4: Attention, Saliency and Grad-CAM Heatmaps for described GRU Ablation



Supplementary Figure 5: Attention, Saliency and Grad-CAM Heatmaps for described Transformer Ablation

